Daneben Projekt: Hardware & Enclosure 3D Design (Detailed)

Technical representation, exact dimensions, and IP67 sealing methodologies.

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1. Orthographic Blueprint & Exact Dimensions

To facilitate real-world manufacturing (e.g., 3D printing or injection molding), the enclosure has been designed with precise measurements. Note: A fully parametric 3D CAD file (`daneben_enclosure.scad`) has been generated in your project folder.

> **Overall Dimensions:**
> - **Width:** 120 mm (140mm including mounting bracket overhangs)
> - **Height:** 150 mm
> - **Depth:** 80 mm
> - **Roof Angle:** 20 Degrees (Houses a 100x60mm 5V Solar Panel)
> - **Wall Thickness:** 3 mm for high impact resistance.

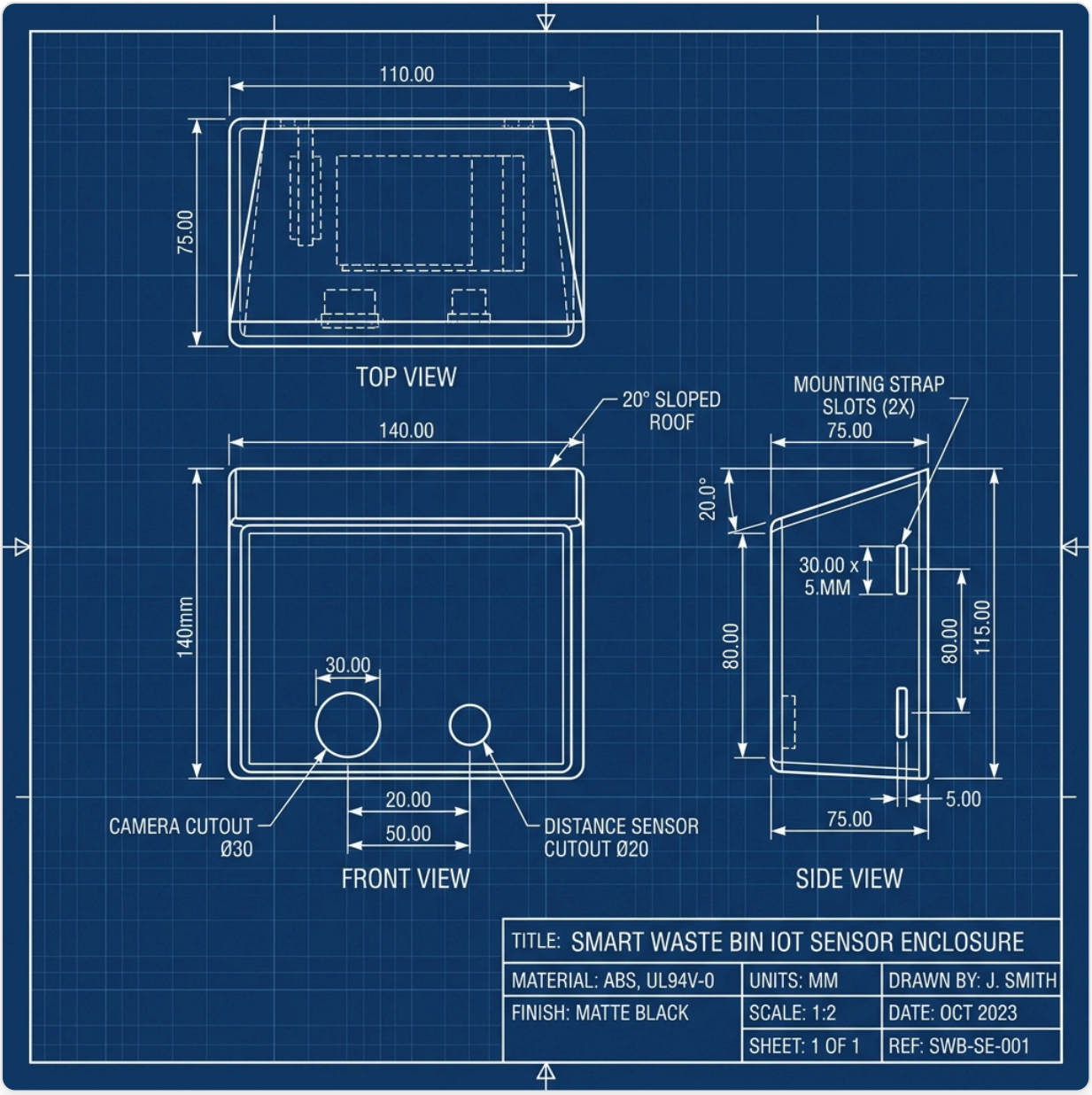


Figure 1: 2D Orthographic projections (Front, Top, and Side views) displaying exact dimensions and cutouts.

2. Waterproofing Mechanisms (Camera & Speaker)

Achieving a true IP67 rating requires specific engineering for the openings:

Step A: Camera Glass Sealing

Problem: The camera lens cannot be exposed to rain, but any covering must not distort the image.

Solution: A 20mm diameter, 2mm thick **borosilicate glass disc** (highly scratch-resistant) is used. It is seated from the *inside* of the enclosure against a flat rubber gasket. A threaded 3D-printed retaining ring screws in behind it, compressing the glass against the gasket. Alternatively, for permanent sealing, the glass is secured using industrial UV-curing waterproof optical adhesive.

Step B: Speaker Audio Vents

Problem: Sound needs to exit the casing, but liquid water must be blocked.

Solution: A series of 3mm holes are drilled into the lower front face. On the *inside* of the housing, a patch of **hydrophobic ePTFE acoustic membrane (like Gore-Tex)** is adhered over the holes using heavy-duty 3M VHB tape. This microscopic mesh allows air pressure and sound waves to pass freely, but the pores are too small for liquid water droplets to penetrate.

Step C: Distance / Presence Sensor

Solution: The distance sensor cutout (16mm) houses a **JSN-SR04T waterproof ultrasonic transducer**. Unlike standard sensors, this transducer is fully potted in metal and is inherently IP67 rated. It pushes through the casing hole and is sealed with a tight-fitting rubber grommet.

3. Waste Bin Attachment Mechanism

The device must be mounted securely to a plastic waste bin without drilling holes in the bin itself (which would void the bin's warranty and let garbage juices leak out).

> ### Mounting Approach:
>
> - **Curved Backplate:** The rear of the enclosure is slightly concave, matching the standard radius of the upper lip/handle of 120L and 240L wheelie bins.
> - **Strap Slots:** Two horizontal slots (75mm wide, 5mm tall) run through the rear mounting block.
> - **Fastening:** Heavy-duty stainless steel band clamps (hose clamps) or industrial nylon zip-ties are threaded through these slots and wrapped tightly around the bin's handle bar or structural ribs. This provides a vice-like grip that cannot be easily vandalized.

4. Step-by-Step Breakdown of the Build

1. **The Base Shell:** This is the main bucket. It features the curved backplate and the strap slots.
2. **Electronics Insertion:** The custom "Inverted T" PCB slides down into rails on the sides of the base shell.
3. **Sensors & Audio:** The waterproof ultrasonic transducer is pushed through its rubber grommet. The camera glass is UV-glued into its recess. The ePTFE membrane is taped over the speaker holes.
4. **Battery Cartridge:** A threaded tubular chamber at the bottom allows a 18650 battery to slide in. A screw-cap with a dedicated O-ring seals it.
5. **The Roof & Solar Panel:** The 20-degree sloped lid holds the 100x60mm solar panel (flush mounted to prevent snagging). A continuous silicone O-ring is laid into the groove of the base shell.
6. **Final Sealing:** The sloped lid is compressed down onto the base shell using four stainless steel M4 hex screws, crushing the O-ring and creating a watertight seal beneath the protective drip edge.



Figure 2: Exterior 3D concept showing the sloped roof, flush solar panel, drip edge, and recessed bottom sensors.

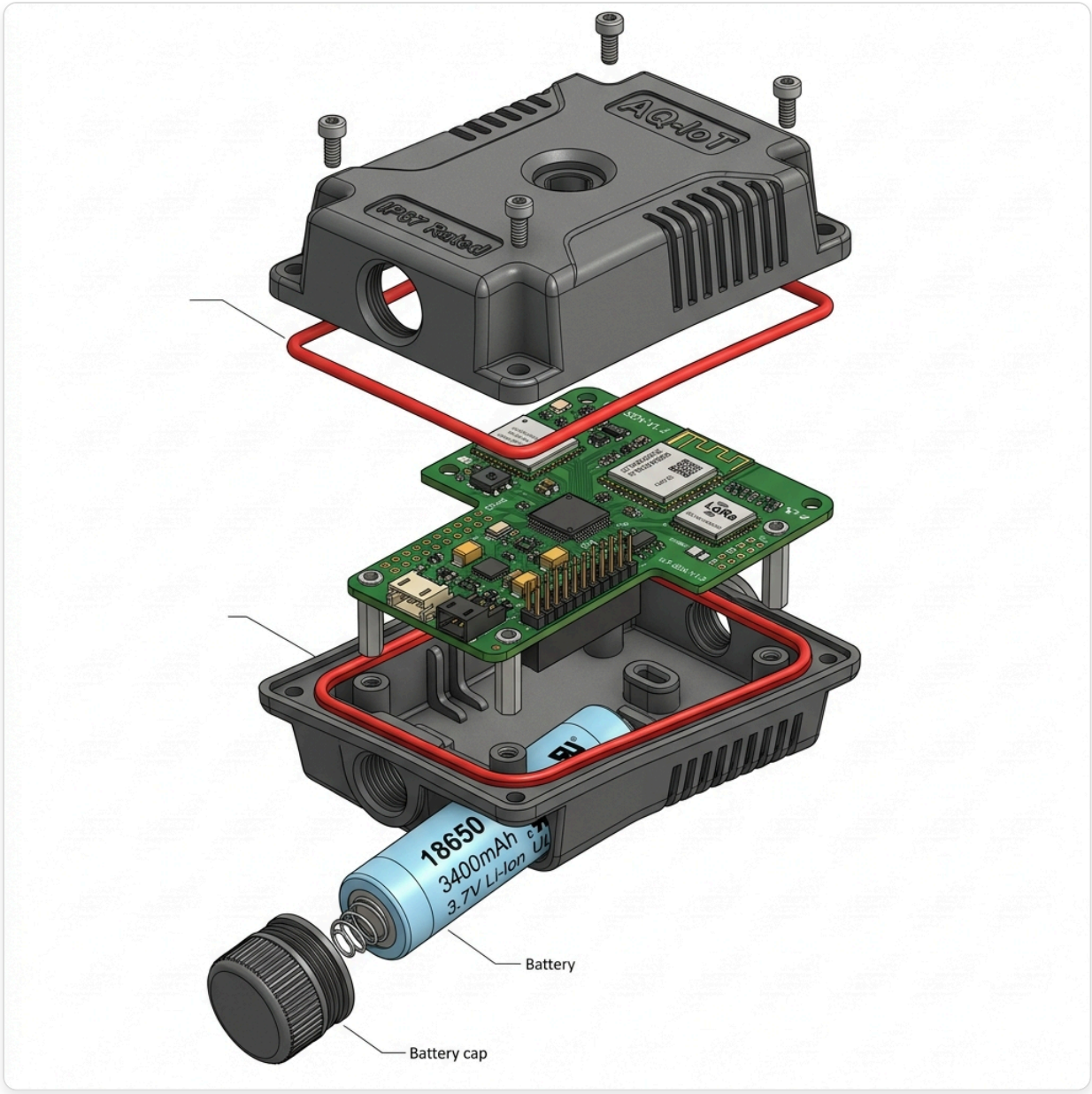


Figure 3: Exploded 3D CAD representation showing the silicone O-rings, custom PCB layout, and the isolated bottom-loading battery cartridge.